KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 9 5.2

Yearly Exam Practice

Practice Exam 1

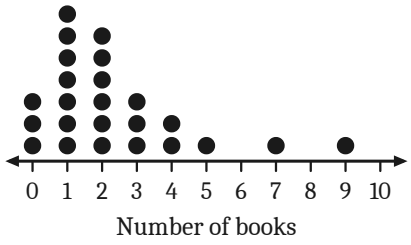
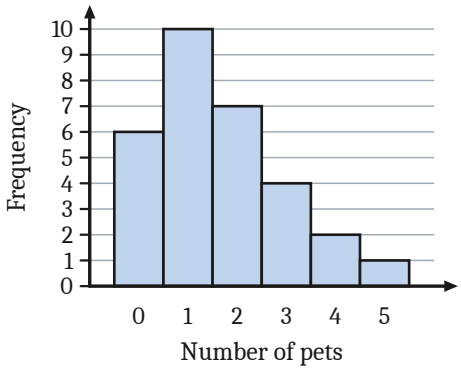
Instructions

1. Working time: 60 minutes
2. Attempt all questions
3. Show ALL working where needed
4. A calculator is permitted
5. Answers are at the back of this booklet

Sections and Marks

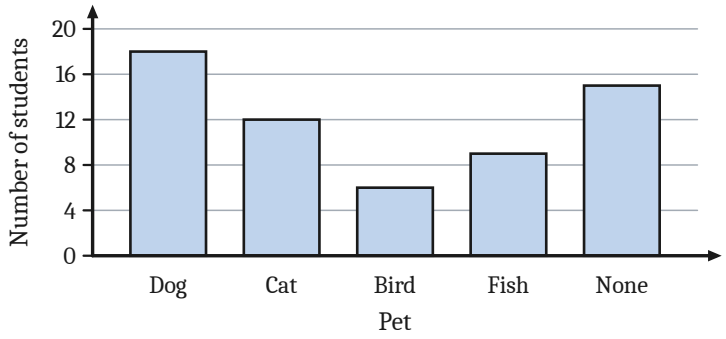
Section 1 - Data	/14
Section 2 - Surface area and volume	/20
Section 3 - Mixed Topics	/30
Total marks	**/64**

Section 1: Data

	Multiple Choice:	Mark
1	Which of the following is **numerical continuous** data? A. The number of brothers and sisters B. Eye colour C. The height of a student D. A rating of a film (poor, fair, good)	1
2	The dot plot shows the number of books read last month of 24 students. Which statement best describes the data? A. Positively skewed, clustered at 1 and 2 B. Negatively skewed, clustered at 1 and 2 C. Symmetrical, clustered at 1 and 2 D. Positively skewed, clustered at 7 and 9 	1
3	The histogram shows the number of pets owned by each of 30 students. What is the mode? A. 5 B. 1 C. 10 D. 2.5 	1
4	A café counted the number of customers it served in each hour of one day: 14, 12, 15, 13, 14, 48, 13, 14 Which measure of centre changes the most if the outlier, 48, is removed? A. The mode B. The median C. The mean D. All three change by the same amount	1

Section 1: Data (continued)

Multiple Choice:

		Mark
5	Students in Year 9 were asked which pet they would most like to own. The column graph shows the results. What percentage of the students chose cat or bird?  A. 18% B. 30% C. 20% D. 36%	1

Short Answer:

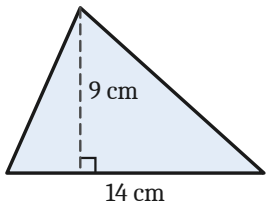
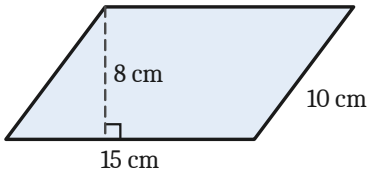
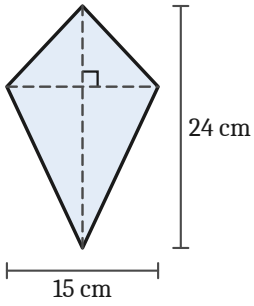
	Question	Working Out	Mark
6	The stem-and-leaf plot shows the marks (out of 70) of 11 students in a science test. Stem | Leaf 2 | 8 3 | 3 5 8 4 | 1 3 6 9 5 | 2 5 6 | 4 Key: 4 | 3 = 43 marks a. Find the mean. b. Find the median. Note: Mean = sum of the scores ÷ number of scores	a. b.	2 1

Section 1: Data (continued)

Worded Problems:

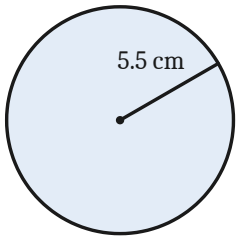
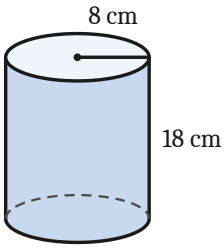
	Question	Working Out	Mark
7	The back-to-back stem-and-leaf plot shows the test marks (out of 100) of 10 students from each of two classes, 9A and 9B. Note: Range = highest score − lowest score	9A | Stem | 9B 3 | 8 4 | 5 8 5 2 | 5 | 1 7 7 6 4 3 1 | 6 | 2 8 2 0 | 7 | 1 7 8 | 3 8 Key: 6 | 3 = 63 marks	
a	Find the median mark for each class.		2
b	Which class usually scored higher? Give a reason.		1
c	Which class is more consistent? Find the range of each class to explain.		3
	Total		**/14**

Section 2: Surface area and volume

	Multiple Choice:	Mark
1	A water tank holds 7.5 kL. How many litres is this? A. 750 L B. 7500 L C. 75 000 L D. 0.0075 L	1
2	What is the area of this triangle? A. 126 cm^2 B. 63 cm^2 C. 23 cm^2 D. 11.5 cm^2 	1
3	What is the area of this parallelogram? Note: Area of a parallelogram: $A = bh$ A. 150 cm^2 B. 60 cm^2 C. 120 cm^2 D. 50 cm^2 	1
4	What is the area of this kite? Note: Area of a kite: $A = \frac{1}{2}xy$ (x and y are the diagonals) A. 180 cm^2 B. 360 cm^2 C. 90 cm^2 D. 39 cm^2 	1

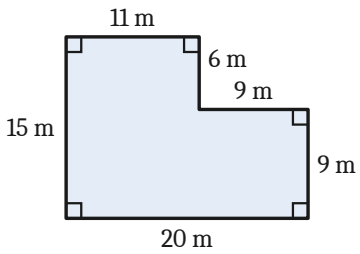
Section 2: Surface area and volume (continued)

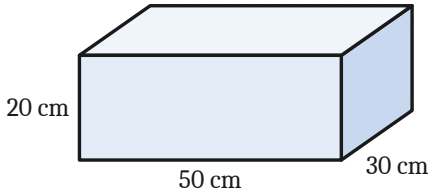
Short Answer:

	Question	Working Out	Mark
5	For this circle, find, correct to 1 decimal place:  a. its circumference b. its area. Note: Circumference: $C = 2\pi r$ Note: Area: $A = \pi r^2$	a. b.	1 1
6	A closed paint tin is a cylinder. It has a radius of 8 cm and a height of 18 cm.  a. Find its volume, correct to 1 decimal place. b. Find its surface area, correct to 1 decimal place. Note: Volume: $V = \pi r^2 h$ Note: Surface area: $SA = 2\pi r^2 + 2\pi rh$	a. b.	2 2

Section 2: Surface area and volume (continued)

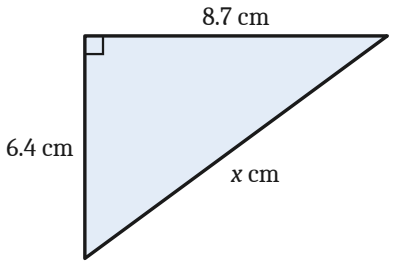
Worded Problems:

	Question	Working Out	Mark
7	The plan shows an L-shaped lawn in a park. All of the corners are right angles. Note: Area of a rectangle: $A = lb$ 		
a	A border is laid around the outside of the lawn. How many metres of border are needed?		1
b	Find the area of the lawn.		3
c	Turf costs $11.60 per square metre. Find the cost of turfing the whole lawn.		1

Section 2: Surface area and volume (continued)			
	Question	**Working Out**	**Mark**
8	A closed toolbox is shaped like a box. It is a rectangular prism 50 cm long, 30 cm wide and 20 cm high. 		
a	Find the volume of the box in cm^3. Note: Volume of a rectangular prism: $V = lwh$		1
b	How many litres does the box hold? Note: 1 L = 1000 cm^3		1
c	The outside of the toolbox is to be painted. Find the surface area of the box. Note: Surface area = the total area of the 6 faces; area of a rectangle: $A = lb$		3
	Total		**/20**

Section 3: Mixed Topics (Chapters 1–8)

Multiple Choice:

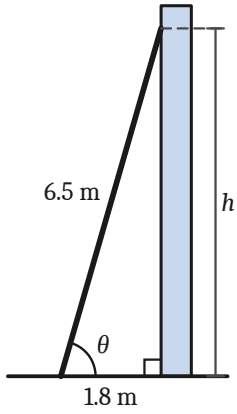
		Mark
1	If $a = 4$ and $b = 7$, evaluate $3a^2 - b$. A. 5 B. 17 C. 41 D. 137	1
2	Find the value of x. Select the closest answer. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ A. 15.1 B. 10.8 C. 5.9 D. 2.3 	1
3	A jacket is marked at \$180. In a sale it is discounted by 25%. What is the selling price? A. \$225.00 B. \$155.00 C. \$135.00 D. \$45.00	1
4	Simplify $5k^4 \times 2k^3$. A. $7k^7$ B. $10k^7$ C. $10k^1$ D. $10k^{12}$	1
5	Priya, a pharmacist, earns an annual salary of \$76 440. How much is this per fortnight? Note: 1 year = 52 weeks = 26 fortnights = 12 months A. \$1470.00 B. \$2940.00 C. \$3185.00 D. \$6370.00	1

Short Answer:

	Question	Working Out	Mark
6	Simplify: a. $6p + 4q - 2p + 3q$ b. $24xy \div 8x$	a. b.	1 1
7	Expand and simplify: $4(2x - 3) - 3(x - 5)$		2

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
8	Factorise: a. $12x + 18$ b. $k^2 - 5k$	a. b.	1 1
9	Noah invests \$6400 for 4 years at 3.5% p.a. simple interest. Calculate the interest earned. Note: Simple interest: $I = Prn$		2
10	A bike shop buys a bike for \$360 and sells it for \$450. a. Find the profit. b. Find the profit as a percentage of the cost price. Note: Percentage profit = profit ÷ cost price × 100%	a. b.	1 1
11	Solve: $6x - 3(x + 2) = x + 14$		3
12	Zara sells solar panels. Zara is paid \$480 per week plus 3% commission on sales. Last week Zara sold \$38 600 worth of solar panels. Calculate Zara's total pay for the week.		2

	Section 3: Mixed Topics (Chapters 1–8) (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
13	Kai works at a supermarket. The normal rate of pay is \$27.60 per hour for the first 38 hours each week, and time-and-a-half for each hour after that. Last week Kai worked 43 hours.		
a	Calculate the pay for the first 38 hours.		1
b	Calculate the total pay for the 43 hours.		2
14	A phone plan costs a \$25 connection fee plus \$40 per month. The total cost, C dollars, for m months is given by the formula $\boldsymbol{C = 25 + 40m}$		
a	Find the total cost for 9 months.		1
b	Sam paid \$505 altogether. Write an equation and solve it to find how many months Sam was on the plan.		2

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
15	A ladder 6.5 m long leans against a vertical wall. The foot of the ladder is 1.8 m from the wall, on level ground. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$ 		
a	Find h, the height the ladder reaches up the wall, correct to 2 decimal places.		2
b	Find the angle θ that the ladder makes with the ground, correct to the nearest degree.		2
	Total		**/30**

Answers

Mark your own work. If an answer is wrong, read the help sheet topic in the last column, then try the question again.

Section 1 answers – Data			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	C (The height of a student)	1	D1 Types of data
2	A (Positively skewed, clustered at 1 and 2)	1	D2 Shape of a distribution
3	B (1)	1	D3 Reading graphs
4	C (The mean)	1	D5 Outliers
5	B (30%)	1	D3 Reading graphs
6a	44 marks	2	D4 Mean, median, mode, range
6b	43 marks	1	D4 Mean, median, mode, range
7a	9A: 63.5; 9B: 65	2	D6 Comparing two data sets
7b	9B: the median for 9B (65) is higher than the median for 9A (63.5).	1	D6 Comparing two data sets
7c	9A: the range is 20 (72 – 52), much smaller than the range for 9B, 50 (88 – 38).	3	D6 Comparing two data sets
	Total	**14**	

Section 2 answers – Surface area and volume			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	B (7500 L)	1	M1 Units
2	B (63 cm^2)	1	M2 Area of shapes
3	C (120 cm^2)	1	M2 Area of shapes
4	A (180 cm^2)	1	M2 Area of shapes
5a	34.6 cm	1	M3 Circles
5b	95.0 cm^2	1	M3 Circles
6a	3619.1 cm^3	2	M5 Volume
6b	1306.9 cm^2	2	M6 Surface area
7a	70 m	1	M4 Composite shapes
7b	246 m^2	3	M4 Composite shapes
7c	$2853.60	1	M4 Composite shapes
8a	30 000 cm^3	1	M5 Volume
8b	30 L	1	M1 Units
8c	6200 cm^2	3	M6 Surface area
	Total	**20**	

Section 3 answers – Mixed Topics (Chapters 1–8)			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	C (41)	1	A1 Substitution
2	B (10.8)	1	T1 Pythagoras
3	C ($135.00)	1	F1 Percentages and profit
4	B ($10k^7$)	1	A5 Index laws
5	B ($2940.00)	1	F3 Earning money
6a	$4p + 7q$	1	A2 Simplifying
6b	$3y$	1	A2 Simplifying
7	$5x + 3$	2	A3 Expanding
8a	$6(2x + 3)$	1	A4 Factorising
8b	$k(k - 5)$	1	A4 Factorising
9	$896.00	2	F2 Simple interest
10a	$90	1	F1 Percentages and profit
10b	25%	1	F1 Percentages and profit
11	$x = 10$	3	E1 Solving equations
12	$1638	2	F3 Earning money
13a	$1048.80	1	F3 Earning money
13b	$1255.80	2	F3 Earning money
14a	$385	1	E2 Formulas and word problems
14b	12 months	2	E2 Formulas and word problems
15a	6.25 m	2	T1 Pythagoras
15b	74°	2	T2 Trigonometry
	Total	**30**	